Ujjwal Kumar Upadhyay

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Research Keywords: Cosmology, Dark energy, Large-scale structure of the universe.

Research Profile

I am a theoretical and statistical cosmologist working at the interface of cosmological theory, Bayesian inference, and observations. My research focuses on developing rigorous statistical methods for cosmological inference and using them to investigate cosmological tensions and anomalies, with particular emphasis on distinguishing signatures of new physics from observational and analysis systematics. My PhD thesis developed a Bayesian errors-in-variables framework for joint inference of galaxy peculiar motions and cosmological parameters from Type Ia supernovae.

Experience

Aug. 2026 – Present **Postdoctoral Visitor**
The Institute of Mathematical Sciences, Chennai, India

Education

2021 – 2026 **PhD in Cosmology**
Indian Institute of Science, Bangalore, India
Raman Research Institute, Bangalore, India
Thesis: *A Bayesian Framework for Joint Inference of Peculiar Motion and Cosmology from Supernovae Observations*
Advisors: Dr. Tarun Deep Saini & Prof. Shiv K. Sethi

2017 – 2019 **MSc, Masters Degree in Physics**
Banaras Hindu University, Varanasi, India
Specialization: Spectroscopy
Thesis: *Phase Space Visualization of Quantum States*
Advisor: Dr. Devendra Kumar Mishra

2014 – 2017 **BSc, Bachelors Degree in Physics**
St. Xavier College, Ranchi, India
Subjects: Physics, Mathematics & Computer Science with `C/C++`

Publications

Accepted for Publication:

[1] Yashi Tiwari, **Ujjwal Upadhyay**, Shao-Jiang Wang and Vivian Poulin, *Can Distance Duality Violation Save Late-time Solutions to the Hubble Tension?*, [arXiv:2607.08654] (**PRD**)

Published:

[2] Ujjwal Upadhyay, Tarun Deep Saini and Shiv K. Sethi, *Bayesian Inference of Peculiar Motions of Low Redshift Galaxies from Type Ia Supernovae Observations*, 10.1093/mnras/stag1752, [arXiv:2603.06469]

[3] Ujjwal Upadhyay, Yashi Tiwari, Tarun Souradeep, *Updated Constraints on Infrared Cutoff Models and Implications for Large-Scale CMB Anomalies*, JCAP06(2026)019, [arXiv:2602.16659]

[4] Brijesh Kanodia, **Ujjwal Upadhyay**, Yashi Tiwari, *Revisiting Cosmic Distance Duality with Megamasers and DESI DR2: Model Independent Constraints on Early–Late Calibration*, Phys.Rev.D 113, 023505, [arXiv:2507.11518]

[5] Ujjwal Upadhyay, Tarun Deep Saini and Shiv K. Sethi, *A Bayesian estimator for peculiar velocity correction in cosmological inference from supernovae data*, 10.1093/mnras/stag381, [arXiv:2502.09258]

[6] Yashi Tiwari, **Ujjwal Upadhyay**, Rajeev Kumar Jain, *Exploring cosmological imprints of phantom crossing with dynamical dark energy in Horndeski gravity*, PhysRevD.111.043530, [arXiv:2412.00931]

Computational Experience

Boltzmann Codes: `class, hi_class`
Experience with Boltzmann codes to study scalar field models of dark matter and dark energy.

Statistical Codes: `Cobaya, Monte Python`
Skilled in Bayesian analysis of cosmological models, with experience in Cobaya and Monte Python codes for statistical inference.

High Performance Computing
Experience with high-performance computing clusters for MCMC analysis.

Statistics & Machine Learning
Bayesian Inference, Model Selection, Non-linear Regression with Errors-in-variables, Gaussian Process Regression, Symbolic Regression, Genetic Algorithm.

Programming Languages
Skilled in C/C++ and Python.

Operating Systems
Proficient in Linux, macOS, and Windows environments.

Other Software & Tools
Proficient in Mathematica, GetDist, and Latex.

Mentoring & Teaching Experience

Mentored, a third-year undergraduate student from IISER Pune on a summer project (2 credits) titled *"Examining the Expansion of the Universe with Supernovae Cosmography"*, from April - August, 2026 (jointly with Dr. Yashi Tiwari).

Mentored, a first-year PhD student on a project titled *"Cosmological Parameter Estimation*

Using Bayesian Statistics and Markov Chain Monte Carlo Techniques", conducted as part of *Astronomical Techniques* course (2 credits), from January - April, 2025 (jointly with Prof. Rajeev Kumar Jain).

Teaching Assistant, for the course 'General Relativity and Cosmology' taught by Dr. Sanved Kolekar, from January 2024 - April 2024 at IISc, Bangalore, India.

Teaching Assistant, for the course 'Fundamentals of Astrophysics' taught by Prof. Nirupam Roy, from August 2023 - December 2023 at IISc, Bangalore, India.

Academic Duties & Outreach

Referee Task
Referee for *Physical Review D* (4), *Physics of the Dark Universe* (1).

Organizational Tasks

2024 – 2026: Organizer of the Cosmology Journal Club at IISc.
2024 – Student volunteer in 42nd meeting of Astronomical Society of India, held at IISc.

IISc Open Day

2026 – Public talk on 'Seeing the Unseen: Recent Advances in Black Hole Observations'.
2025 – Public talk on 'Decoding the Cosmos: How the Universe Evolves'.
2024 – Poster presentation on 'The Standard Model of Cosmology'.
2023 – Public talk on 'Multi-messenger Astronomy'.

Awards & Achievements

Awarded ITS grant by Anusandhan National Research Foundation (ANRF) in July 2026 to present research work in an international conference in the Netherlands.

Best Teaching Assistant Award 2024 for the course Fundamentals of Astrophysics, Indian Institute of Science, Bangalore.

All India Rank 28 in the Graduate Aptitude Test for Engineering (GATE) for Physics, 2021.

All India Rank 48 in Council for Scientific and Industrial Research (CSIR) – Lectureship for Physics, 2021.

State Topper in the National Graduate Physics Examination (NGPE), 2017.

Invited Talks

[05/2026] – **Invited Talk** in the Department of Astronomy and Astrophysics at the Tata Institute of Fundamental Research, Mumbai, India.

[01/2025] – **Invited Talk** in the PDA student seminar at the Indian Institute of Science, Bangalore.

Selected Conference & Seminars

[11/2026] – **Oral Presentation** in the 6th BRICS-AGAC Symposium, in Hangzhou, China.

[09/2026] – **Oral Presentation** in Cosmic Connections Second Edition: AI/ML Across Cosmic Scales at the Institute of Mathematical Sciences, Chennai, India.

[12/2025] – **Oral Presentation** in the 3rd Neighbourhood Cosmology Meeting at the Indian Institute of Astrophysics, Bangalore.

[07/2025] – **Oral Presentation** in Cosmic Connections: Bridging the Early and Late Universe at the Institute of Mathematical Sciences, Chennai, India.

[06/2025] – **Oral Presentation** in Cosmology from Home 2025 (Online).

[11/2024] – **Poster Presentation** at Kashiwa-no-ha Dark Matter and Cosmology Symposium, Kavli IPMU, Tokyo, University.

[10/2024] – **Poster Presentation** at 27th International Conference on Particle Physics and Cosmology, COSMO'24, YITP, Kyoto University, Japan.

[08/2024] – **Poster Presentation** in Frontiers of Particle Physics Conference, Centre of High Energy Physics, IISc.

[04/2024] – **Oral Presentation** in In-house Symposium at the Raman Research Institute, Bangalore, India.

[12/2023] – **Oral presentation** in 10th International Conference on Gravitation and Cosmology (ICGC) at Indian Institute of Technology (IIT), Guwahati, India.

November, 2026